

Instructions for Participant

Study: Evaluating AI Guardrails and Their Effect on Students' Independent Critical Thinking

Purpose of the Study

This study explores how different ways of using AI tools affect learning, reasoning, and problem-solving ability.

You will be assigned to one of three groups:

- **Group A: Unrestricted AI**
- **Group B: AI with Guardrails**
- **Group C: No AI**

Study Flow

You will complete the following steps:

1. **Pre-Test (No AI)**
2. **Learning Phase (based on your assigned group)**
3. **Post-Test (No AI)**
4. **Survey & Reflection**

General Guidelines

- Answer all questions to the best of your ability
- Provide explanations wherever required
- Do not collaborate with others
- Follow instructions specific to your group

Group-Specific Instructions

Group A: Unrestricted AI

You are allowed to use AI tools (e.g., ChatGPT, Gemini) freely during the learning phase.

Guidelines:

- Use AI in any way you prefer
- You may ask for answers, explanations, or examples
- There are no restrictions on usage

Group B: AI with Guardrails

You are allowed to use AI tools, but you must follow the **guardrail steps**:

Required Steps:

1. **Ask AI**
Use AI to understand the problem
2. **Verify Output**
 - Cross-check answers
 - Ask follow-up questions
 - Ensure correctness
3. **Explain in Your Own Words**
 - Do not copy AI responses
 - Write your own explanation
4. **Reflect**
 - Think about what you learned
 - Consider if you can solve it without AI

Group C: No AI

You are **not allowed to use any AI tools** during the learning phase.

Guidelines:

- Solve problems independently
- Use your own knowledge and reasoning
- External AI tools are not permitted

Important Rules

- During **Pre-Test and Post-Test**, **NO AI is allowed for any group**
- Violating this rule may affect the study results

Confidentiality

- Your responses will remain anonymous
- Data will be used only for research purposes

Final Note

This study is not about getting the highest score.

It is about understanding **how you think and learn**.

Please participate honestly and thoughtfully.